

Running DDSCAT

A Beginner's Guide

Sophia Kugler
Summer Student Internship
Christian-Albrechts-Universität zu Kiel

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1 Introduction

Dust grains absorb, scatter, and re-emit light and therefore influence the observed radiation in many astrophysical environments. Their optical properties depend on their size, material composition, internal structure, shape, and on the wavelength of the incident light.

DDSCAT is a program for calculating absorption and scattering by such particles. It was developed by B. T. Draine and P. J. Flatau and uses the discrete dipole approximation (DDA). In the DDA, a continuous particle is replaced by a finite number of polarizable points, called dipoles.

The purpose of this manual is to provide a practical introduction to DDSCAT for users who are not familiar with the program. The focus is on the files and parameters that are required to set up and run a calculation, not on the internal numerical implementation. This manual uses DDSCAT 7.3.4.

2 DDSCAT

2.1 Installation and directory structure

DDSCAT 7.3.4 and the corresponding User Guide can be downloaded from

<https://ddscat.wikidot.com/downloads>

The source distribution contains several directories. A simplified directory structure can look like this:

```
1 DDA/
2 |-- src/
3 |-- examples_exp/
4 '-- doc/
```

The directory `src` contains the Fortran source code. The compiled DDSCAT executable is also created there. The directory `examples_exp` contains example calculations and can be used to store new simulation directories. The directory `doc` contains additional documentation and is not needed to run simulations.

2.1.1 Compiling DDSCAT

DDSCAT is written in Fortran. On Linux, the GNU Fortran compiler `gfortran` can be used.

It can be checked with

```
1 gfortran --version
```

To compile DDSCAT, go to the source directory,

```
1 cd ~/Downloads/DDA/src
```

and run

```
1 make ddscat
```

The command `make` reads the Makefile supplied with DDSCAT and uses the Fortran compiler to compile and link the required source files.

After successful compilation, the executable is located at, for example,

```
1 ~/Downloads/DDA/src/ddscat
```

The exact path depends on where DDSCAT was installed.

The executable itself does not contain the physical setup of a calculation. The particle, wavelengths, material, and numerical settings are specified separately through input files.

The automated workflow described below additionally requires Python 3.11 or newer. The plotting scripts use NumPy and Matplotlib. The installed Python version can be checked with

```
1 python3 --version
```

2.2 Files required for a calculation

The most important input file is

```
1 ddscat.par
```

This parameter file tells DDSCAT what should be calculated.

A material file containing the wavelength-dependent optical constants is also required.

For a built-in target, such as an ellipsoid, these two files are sufficient:

```
1 run_directory/
2 |-- ddscat.par
3 '-- diel.dat
```

For a custom target geometry, a third file is required:

```
1 run_directory/
2 |-- ddscat.par
3 |-- diel.dat
4 '-- shape.dat
```

2.3 Dust material and optical constants

The optical properties of the dust material are described by the complex refractive index

$$m(\lambda) = n(\lambda) + ik(\lambda). \quad (1)$$

Here, n is the real part of the refractive index and k is the imaginary part. Both quantities normally depend on wavelength.

DDSCAT reads these quantities from a text file. A simplified material file may look like

```
1 Astronomical silicate
2 1 2 3 0 0 = columns for wave, Re(n), Im(n), eps1, eps2
3 wave(um) Re(n) Im(n)
4 0.3000 1.70 0.030
5 0.4000 1.68 0.025
6 0.5000 1.66 0.020
```

The first line is a description of the material.

The second line tells DDSCAT which columns contain the wavelength, n , and k .

The numerical data then follow. The wavelength range of the material file has to cover the wavelength range used in the simulation.

In a run directory, it is useful to copy the selected material file to a short name such as

```
1 diel.dat
```

and refer to this file from `ddscat.par`.

2.4 The parameter file `ddscat.par`

The file `ddscat.par` controls the complete DDSCAT calculation. DDSCAT searches for a file with exactly this name in the current working directory.

A complete example for a compact homogeneous sphere is shown below. The following sections explain the individual lines.

```
1  ' ===== Parameter file for DDSCAT 7.3 ===== '
2
3  '***** Preliminaries *****'
4  'NOTORQ' = CMDTRQ
5  'PBCGS2' = CMDSOL
6  'GPFAFT' = CMDFFT
7  'GKDLDR' = CALPHA
8  'NOTBIN' = CBINFLAG
9
10 '***** Initial Memory Allocation *****'
11 100 100 100
12
13 '***** Target Geometry and Composition *****'
14 'ELLIPSOID' = CSHAPE
15 70 70 70
16 1 = NCOMP
17 'diel.dat'
18
19 '***** Additional Nearfield calculation? *****'
20 0 = NRFLD
21 0.0 0.0 0.0 0.0 0.0 0.0
22
23 '***** Error Tolerance *****'
24 1.0e-5 = TOL
25
26 '***** Maximum number of iterations *****'
27 1000 = MXITER
28
29 '***** Interaction cutoff parameter *****'
30 1.0e-2 = GAMMA
31
32 '***** Angular resolution *****'
33 0.5 = ETASCA
34
35 '***** Vacuum wavelengths (micron) *****'
36 0.3 100.0 60 'LOG'
37
38 '***** Refractive index of ambient medium *****'
39 1.0 = NAMBIENT
```

```

40 |
41 | '**** Effective Radii (micron) ****'
42 | 1.0 1.0 1 'LIN'
43 |
44 | '**** Define Incident Polarizations ****'
45 | (0,0) (1.,0.) (0.,0.)
46 | 2 = IORTH
47 |
48 | '**** Specify which output files to write ****'
49 | 0 = IWRKSC
50 |
51 | '**** Prescribe Target Rotations ****'
52 | 0.0 0.0 1 = BETAMI BETAMX NBETA
53 | 0.0 0.0 1 = THETMI THETMX NTHETA
54 | 0.0 0.0 1 = PHIMIN PHIMAX NPFI
55 |
56 | '**** Specify first I WAV, IRAD, IORI ****'
57 | 0 0 0
58 |
59 | '**** Select Elements of S_ij Matrix to Print ****'
60 | 6 = NSMELTS
61 | 11 12 21 22 31 41
62 |
63 | '**** Specify Scattered Directions ****'
64 | 'LFRAME'
65 | 1
66 | 0.0 0.0 180.0 5.0

```

2.4.1 Preliminaries

The first block selects several numerical methods:

```
1 'NOTORQ' = CMDTRQ
```

NOTORQ means that radiation torques are not calculated. This is sufficient when the main quantities of interest are absorption and scattering.

```
1 'PBCGS2' = CMDSOL
```

This selects the iterative solver. DDSCAT has to solve a large system of equations for the polarization of all dipoles. PBCGS2 is one of the available iterative methods.

```
1 'GPFAFT' = CMDFFT
```

This selects the Fast Fourier Transform method used to speed up the interaction calculations between the dipoles.

```
1 'GKDLDR' = CALPHA
```

This selects the prescription used to calculate the polarizability of the individual dipoles from the optical properties of the material.

```
1 'NOTBIN' = CBINFLAG
```

This disables additional binary output. For most basic calculations, the normal text output is sufficient.

Initial memory allocation

```
1 100 100 100
```

These three numbers specify the dimensions available for constructing the target on the dipole lattice.

They do not mean that all 100^3 positions are occupied by dipoles. They only define a sufficiently large grid in which the target can be generated.

For example, a target with a diameter of approximately 70 dipole spacings fits inside a $100 \times 100 \times 100$ allocation.

Target geometry

For a homogeneous sphere, a built-in ellipsoid can be used:

```
1 'ELLIPSOID' = CSHAPE
2 70 70 70
```

The three numbers give the ellipsoid dimensions in units of the dipole spacing d .

Because all three values are equal, the target is a sphere.

Thus,

$$\frac{D}{d} \approx 70,$$

where D is the diameter of the target.

Increasing these values represents the same geometrical shape with more dipoles and therefore increases the numerical resolution.

The physical size of the grain is *not* determined by these three numbers. The physical size is specified later through the effective radius.

Number of materials

```
1 1 = NCOMP
2 'diel.dat'
```

NCOMP is the number of different dielectric materials in the target.

For a homogeneous grain,

$$N_{\text{COMP}} = 1.$$

The following line gives the file containing the optical constants of this material.

For a target consisting of several materials, several dielectric files can be supplied.

Near-field calculation

```
1 0 = NRFLD
2 0.0 0.0 0.0 0.0 0.0 0.0
```

Setting **NRFLD** to zero switches off the additional near-field calculation.

Near-field calculations are useful when the electric and magnetic fields close to the target are needed. They require additional computation and are not necessary for a standard calculation of Q_{abs} and Q_{sca} .

The following six numbers define the extension of the near-field volume and are irrelevant when **NRFLD**=0.

Solver tolerance and maximum iterations

```
1 1.0e-5 = TOL
2 1000 = MXITER
```

TOL gives the requested convergence tolerance of the iterative solver.

Here, the solver tries to reduce its error below approximately

$$10^{-5}.$$

MXITER is the maximum number of iterations.

If **DDSCAT** reaches 1000 iterations without reaching the requested tolerance, the calculation may stop with an error such as

```
1 FATAL ERROR IN PROCEDURE: ZBCG2WP
2 ITERN>ITERMX
```

This means that the iterative solver did not converge for the selected combination of particle, material, wavelength, and resolution.

Interaction and angular parameters

```
1 1.0e-2 = GAMMA
2 0.5 = ETASCA
```

GAMMA is a numerical parameter used in the treatment of the dipole interaction.

ETASCA controls the angular resolution used when **DDSCAT** integrates scattering quantities over direction.

For first calculations, these values can normally be kept at their standard settings.

Wavelength range

```
1 0.3 100.0 60 'LOG'
```

This line means

$$\lambda_{\text{min}} = 0.3 \mu\text{m},$$

$$\lambda_{\text{max}} = 100 \mu\text{m},$$

with 60 wavelength values.

The final entry determines how the wavelength points are distributed.

- LIN: equally spaced in wavelength,
- LOG: equally spaced in logarithmic wavelength,
- INV: equally spaced in inverse wavelength.

A logarithmic grid is useful when a large wavelength interval is studied.

Ambient medium

```
1 1.0 = NAMBIENT
```

This is the refractive index of the medium surrounding the particle.

For a grain in vacuum,

$$n_{\text{ambient}} = 1.$$

Effective radius

```
1 1.0 1.0 1 'LIN'
```

These entries have the form

```
1 minimum_radius maximum_radius number_of_radai spacing
```

The example therefore calculates one effective radius,

$$a_{\text{eff}} = 1 \mu\text{m}.$$

DDSCAT defines the effective radius as

$$a_{\text{eff}} = \left(\frac{3V_{\text{material}}}{4\pi} \right)^{1/3}. \quad (2)$$

It is the radius of a sphere having the same material volume as the DDSCAT target.

For a compact sphere,

$$a_{\text{eff}} = a.$$

For porous or non-spherical targets, the meaning of a_{eff} should be considered carefully because it is determined from material volume and does not necessarily correspond to an outer radius.

Incident polarization

```
1 (0,0) (1.,0.) (0.,0.)
2 2 = IORTH
```

The first line specifies the polarization vector of the incident electromagnetic wave.

IORTH=2 tells DDSCAT to calculate two orthogonal incident polarizations.

For a perfectly homogeneous sphere, the result does not depend on the orientation of the polarization. For non-spherical particles, the orientation can become important.

Output option

```
1 0 = IWRKSC
```

This controls whether additional working files from the scattering calculation are written.

Setting the value to zero avoids unnecessary files for a standard run.

Target orientation

```
1 0.0 0.0 1 = BETAMI BETAMX NBETA
2 0.0 0.0 1 = THETMI THETMX NTHETA
3 0.0 0.0 1 = PHIMIN PHIMAX NPHI
```

The orientation of a DDSCAT target relative to the incident light is described by three angles, usually denoted by β , Θ , and Φ .

Each line specifies

```
1 minimum_angle maximum_angle number_of_values
```

In this example, only one orientation is calculated.

For a sphere, the orientation does not affect the result. For an ellipsoid, aggregate, or irregular particle, several orientations may have to be calculated or averaged.

Starting indices

```
1 0 0 0
```

These are the starting indices for wavelength, radius, and orientation.

For a new calculation, all three values are normally zero.

They can be useful when restarting particular parts of a calculation.

Scattering matrix elements

```
1 6 = NSMELTS
2 11 12 21 22 31 41
```

NSMELTS specifies how many Mueller scattering-matrix elements should be written.

The second line selects the elements.

For example,

$$S_{11}$$

describes the angular distribution of scattered intensity and is one of the most commonly used scattering-matrix elements.

Scattering directions

```
1 'LFRAME '
2 1
3 0.0 0.0 180.0 5.0
```

LFRAME means that scattering directions are specified in the laboratory frame.

The value 1 means that one scattering plane is requested.

The final line requests scattering angles from 0° to 180° in steps of 5°. These settings are mainly relevant when angular scattering or the Mueller matrix is analysed.

2.5 Custom geometries and shape.dat

Instead of a built-in DDSCAT shape, a custom geometry can be read from a file.

The corresponding target definition is

```
1 'FROM_FILE' = CSHAPE
2 no SHPAR parameters needed
```

The working directory must then contain a file called

```
1 shape.dat
```

A simplified example is

```
1 Example target
2 4 = NAT
3 1.0 0.0 0.0 = A1
4 0.0 1.0 0.0 = A2
5 1.0 1.0 1.0 = lattice spacings
6 0.5 0.5 0.5 = X0
7 JA IX IY IZ ICOMPX ICOMPY ICOMPZ
8 1 0 0 0 1 1 1
9 2 1 0 0 1 1 1
10 3 0 1 0 1 1 1
11 4 0 0 1 1 1 1
```

NAT is the number of occupied dipoles.

The columns IX, IY, and IZ contain their positions on the cubic lattice.

The final three columns specify which dielectric material is used for each polarization direction.

For a homogeneous isotropic target made from material 1, these entries are

```
1 1 1 1
```

for every dipole.

This approach can be used to create porous grains by removing selected dipoles from a compact target or to import more complicated geometries.

2.6 Running DDSCAT

DDSCAT has to be started from the directory containing `ddscat.par`.

For example,

```
1 cd ~/Downloads/DDA/examples_exp/MY_RUN
2 ~/Downloads/DDA/src/ddscat
```

DDSCAT automatically searches for

```
1 ddscat.par
```

in the current directory.

During the calculation, information about the current wavelength and solver convergence is written to the terminal and to log files.

A successful calculation ends with a message similar to

```
1 DDSCAT normal termination
```

2.7 DDSCAT output files

DDSCAT produces several output files. The exact number depends on the number of wavelengths, radii, orientations, and selected output options.

The most important files for a first analysis are summarized below.

Table 1: Important DDSCAT output files.

File	Content
<code>qtable</code>	main wavelength-dependent optical quantities
<code>qtable2</code>	additional wavelength-dependent quantities
<code>wXXXrXXX.avg</code>	detailed averaged result for one wavelength and radius
<code>wXXXrXXXkXXX.sca</code>	angular scattering information
<code>target.out</code>	information about the target geometry
<code>ddscat.log_000</code>	numerical information and solver convergence

2.7.1 qtable

For most basic wavelength-dependent analyses, the most useful file is

```
1 qtable
```

The main columns are

$$a_{\text{eff}}, \quad \lambda, \quad Q_{\text{ext}}, \quad Q_{\text{abs}}, \quad Q_{\text{sca}}, \quad g.$$

The efficiencies satisfy

$$Q_{\text{ext}} = Q_{\text{abs}} + Q_{\text{sca}}. \quad (3)$$

The efficiency factors are dimensionless and are related to the corresponding cross sections by $Q = \frac{C}{\pi a_{\text{eff}}^2}$, where the effective radius is defined as $a_{\text{eff}} = \left(\frac{3V_{\text{mat}}}{4\pi}\right)^{\frac{1}{3}}$

The asymmetry parameter $g = \langle \cos \theta \rangle$ describes the preferred scattering direction. Values close to zero indicate approximately symmetric scattering, while positive values indicate preferential forward scattering.

2.7.2 .avg files

DDSCAT produces files such as

```
1 w000r000.avg
2 w001r000.avg
3 w002r000.avg
```

The number after **w** specifies the wavelength index and the number after **r** specifies the effective-radius index.

These files contain more detailed information than **qtable**.

2.7.3 .sca files

Files such as

```
1 w000r000k000.sca
```

contain the angular scattering quantities for a selected wavelength, radius, and orientation.

They can be used, for example, to analyse

$$S_{11}(\theta)$$

or polarization-related Mueller-matrix elements.

2.7.4 Checking convergence

The file

```
1 ddscat.log_000
```

is useful if a calculation fails.

A common error is

```
1 FATAL ERROR IN PROCEDURE: ZBCG2WP
2 ITERN > ITERMX
```

This means that the iterative solver reached **MXITER** before satisfying **TOL**.

It is also useful to check the DDA resolution parameter

$$|m|kd, \tag{4}$$

where

$$k = \frac{2\pi}{\lambda}$$

and d is the dipole spacing.

A commonly used approximate condition for integrated absorption and scattering cross sections is

$$|m|kd < 1.$$

Smaller values correspond to a better electromagnetic resolution.

3 Automated workflow

Writing `ddscat.par` manually is useful when learning DDSCAT, but it becomes inconvenient when many calculations are performed.

The accompanying GitHub repository therefore uses one configuration file containing the parameters that a user is expected to change. The scripts and an example input file are available at

<https://github.com/sckugler/ddscat-for-beginners>

The workflow is

`input.toml` → `generate_ddscat.py` → `ddscat.par` → `main.sh` → DDSCAT.

The user normally only edits

```
1 input.toml
```

while DDSCAT itself is treated as a black box.

3.1 The input file

A shortened example is

```
1 [paths]
2
3 # compiled ddscat executable
4 ddscat_executable = "~/DDA/src/ddscat"
5
6 # directory for the calculation
7 run_directory = "./runs/solid_sphere"
8
9 # optical constants
10 material_file = "~/DDA/diel/astrosil"
11
12
13 [target]
14
15 # ELLIPSOID for a built-in ellipsoid
16 shape = "ELLIPSOID"
17
18 # target dimensions in dipole units
19 shape_parameters = [70, 70, 70]
20
21
22 [wavelength]
23
24 # wavelength interval in micrometres
25 minimum_um = 0.3
26 maximum_um = 100.0
27 count = 60
28 spacing = "LOG"
29
30
```

```

31 [effective_radius]
32
33 # effective radius in micrometres
34 minimum_um = 1.0
35 maximum_um = 1.0
36 count = 1
37 spacing = "LIN"
38
39
40 [numerics]
41
42 # solver settings
43 solver = "PBCGS2"
44 tolerance = 1.0e-5
45 max_iterations = 1000

```

After editing this file, the DDSCAT input can be generated with

```
1 python3 generate_ddscat.py input.toml
```

The script creates the run directory, writes `ddscat.par`, and copies the required material and geometry files into the run directory.

3.2 Running on the cluster

The simulation is submitted through

```
1 main.sh
```

For a Slurm cluster, its beginning may look like

```

1 #!/bin/bash
2
3 #SBATCH --job-name=ddscat
4 #SBATCH --partition=student-l
5 #SBATCH --qos=long
6 #SBATCH --nodes=1
7 #SBATCH --ntasks=1
8 #SBATCH --cpus-per-task=1
9 #SBATCH --time=1-00:00:00
10 #SBATCH --mem=8G

```

The simulation can then be submitted with

```
1 sbatch main.sh input.toml
```

The current status can be checked with

```
1 squeue -u $USER
```

More detailed instructions for using the scripts are given in the `README.md` of the repository.

4 Example: Solid homogeneous sphere

This section demonstrates a complete calculation for a simple test case. The input files are created manually to demonstrate what DDSCAT actually requires. For later calculations, the automated workflow described in the previous section can be used instead. A compact homogeneous sphere is a useful first DDSCAT calculation because the geometry is simple and the result can later be compared with the exact Mie solution.

Step 1: Define the calculation

In this example, the particle is

- a compact sphere,
- made from astronomical silicate,
- with effective radius $a_{\text{eff}} = 1 \mu\text{m}$,
- calculated from 0.3 to $100 \mu\text{m}$,
- represented with approximately $D/d = 70$.

Because the particle is a compact sphere,

$$a_{\text{eff}} = a = 1 \mu\text{m}.$$

Step 2: Create a run directory

First create a new directory.

For example,

```
1 cd ~/Downloads/DDA/examples_exp
2 mkdir SOLID_SPHERE
3 cd SOLID_SPHERE
```

Step 3: Add the material

For this example, astronomical silicate is used.

Copy the material file into the new run directory:

```
1 cp ../diel/astrosil diel.dat
```

The directory now contains

```
1 SOLID_SPHERE/
2 '-- diel.dat
```

Step 4: Create ddscat.par

Create the parameter file,

```
1 nano ddscat.par
```

and use the settings described in Section 2.4.

The important physical parameters for this example are

```
1 'ELLIPSOID' = CSHAPE
2 70 70 70
3 1 = NCOMP
4 'diel.dat'
5
6 0.3 100.0 60 'LOG'
7
8 1.0 1.0 1 'LIN'
```

The equal ellipsoid dimensions define a sphere.

The value 70 sets the approximate number of dipole spacings across its diameter.

The wavelength line requests 60 logarithmically spaced wavelengths between 0.3 and 100 μm .

The effective-radius line sets the physical grain radius to 1 μm .

Step 5: Run DDSCAT

The directory should now contain

```
1 SOLID_SPHERE/
2 |-- ddscat.par
3 '-- diel.dat
```

Run DDSCAT from this directory:

```
1 ~/Downloads/DDA/src/ddscat
```

A successful calculation should eventually print

```
1 DDSCAT normal termination
```

After the run, the directory contains several new files, including

```
1 qtable
2 qtable2
3 target.out
4 ddscat.log_000
5 w000r000.avg
6 w001r000.avg
7 ...
```

Step 6: Check the result

Open the beginning of qtable:

```
1 head -15 qtable
```

Each numerical row corresponds to one wavelength.

For this example, there should be 60 wavelength entries.

The most important quantity for the following example is

$$Q_{\text{abs}}(\lambda).$$

Step 7: Plot the absorption efficiency

The following Python script reads the numerical rows of `qtable` and plots the absorption efficiency.

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4
5 def read_qtable(filename):
6
7     rows = []
8
9     with open(filename) as file:
10
11         for line in file:
12
13             values = line.split()
14
15             if len(values) < 8:
16                 continue
17
18             try:
19                 row = [
20                     float(x)
21                     for x in values[:8]
22                 ]
23
24             except ValueError:
25                 continue
26
27             rows.append(row)
28
29     return np.array(rows)
30
31
32 data = read_qtable(
33     "qtable"
34 )
35
36 wavelength = data[:, 1]
37 qabs = data[:, 3]
38
39
40 plt.figure(
41     figsize=(8, 5.5)
42 )
```

```

43 |
44 | plt.plot(
45 |     wavelength,
46 |     qabs
47 | )
48 |
49 | plt.xscale("log")
50 | plt.yscale("log")
51 |
52 | plt.xlabel(
53 |     r"Wavelength  $\lambda$  [ $\mu\text{m}$ ]"
54 | )
55 |
56 | plt.ylabel(
57 |     r" $Q_{\text{abs}}$ "
58 | )
59 |
60 | plt.title(
61 |     r"Absorption efficiency of a compact Astrosil sphere"
62 | )
63 |
64 | plt.grid(
65 |     alpha=0.3
66 | )
67 |
68 | plt.tight_layout()
69 | plt.show()

```

The resulting plot shows how efficiently the grain absorbs radiation at different wavelengths. For astronomical silicate, a strong spectral feature is expected around

$$\lambda \approx 10 \mu\text{m}.$$

This feature originates from the wavelength-dependent optical properties of the silicate material.

plot

What has been varied and what has been kept fixed?

For this example,

- the material is fixed to astronomical silicate,
- the grain radius is fixed to $1 \mu\text{m}$,
- the shape is fixed to a compact sphere,
- the dipole resolution is fixed,
- only the wavelength is varied.

The resulting curve can therefore be interpreted as the wavelength dependence of the absorption efficiency for one specific grain.

From this simple starting point, further calculations can change one property at a time, for example the grain size, material, porosity, or shape.

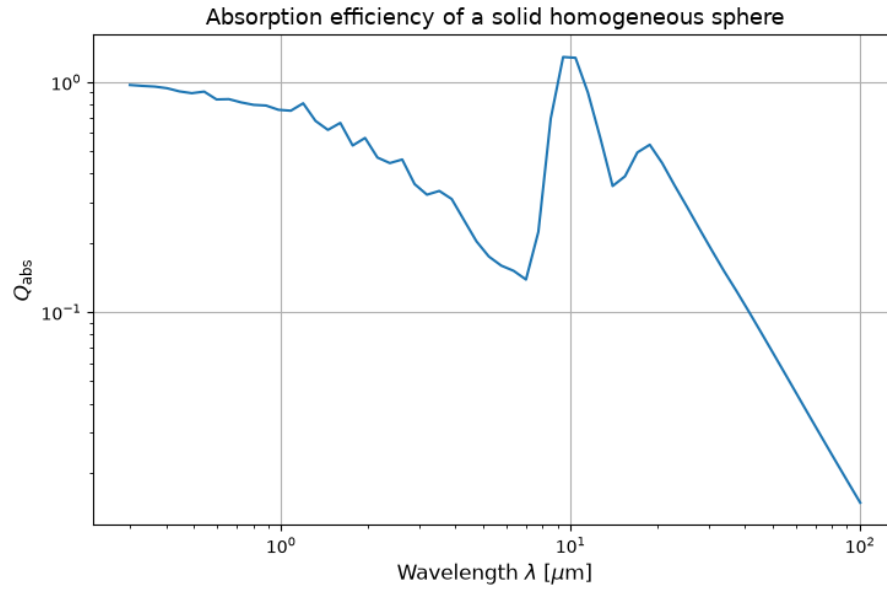


Figure 1: Absorption efficiency of a homogeneous astronomical-silicate sphere with $a_{\text{eff}} = 1 \mu\text{m}$ as a function of wavelength.

References

- Draine, B. T. & Flatau, P. J., DDSCAT 7.3 User Guide.
- Draine, B. T. & Flatau, P. J. (1994), Discrete-dipole approximation for scattering calculations, *Journal of the Optical Society of America A*, 11, 1491.